Experiment No 3

Course Outcome:CO1 **Blooms Level: L4**

Aim: Analyze the inferential statistical results of the given dataset.

Abstract:
The aim of this experiment is to enhance practical understanding of statistical hypothesis testing through realistic data-driven scenarios using appropriate *t-test* techniques.
Given three real-world datasets—(i) the Rice Bag Weight dataset from a retail data mart, (ii) the Pre-Test and Post-Test Score dataset representing student performance before and after studying a learning module, and (iii) the Crocin Tablet Body Temperature dataset recording patient temperatures before and after medication—design and implement programs to perform appropriate t-test–based inferential statistical analyses.

Case 1: One-Sample t-test (Rice Bag Dataset)
Given the Rice Bag Weight dataset collected from a retail data mart, write a program to perform a one-sample t-test to analyze whether the average weight of rice bags supplied differs significantly from the standard population mean weight specified by the vendor. The program should formulate appropriate null and alternative hypotheses, compute the t-statistic and p-value, and interpret the results at a chosen level of significance to determine compliance with the expected weight standard.

Case 2 : Paired Sample t-test (Pre-Test / Post-Test Dataset)
Problem Statement:
Given a Pre-Test and Post-Test Score dataset representing the performance of the same group of students before and after studying a learning module, develop a program to perform a paired sample t-test. The objective is to statistically analyze whether the instructional intervention has resulted in a significant improvement in student performance by comparing the mean pre-test and post-test scores using hypothesis testing.

Case 3: Paired Sample t-test (Crocin Tablet Dataset)
Problem Statement:
Given the Crocin Tablet Temperature dataset, which records body temperature readings of patients before and after administering a Crocin tablet, write a program to perform a paired sample t-test to determine whether the medication leads to a statistically significant reduction in body temperature. The program should compute the mean difference, t-value, and p-value, and draw conclusions regarding the effectiveness of the tablet based on inferential statistical analysis.

Theory:

- Z-Test & T-Tests are Parametric Tests, where the Null Hypothesis is less than, greater than or equal to some value.
- A z-test is used if the population variance is known, or if the sample size is larger than 30.
- If the sample size is less than 30 and the population variance is unknown, we must use a t-test.

T test is a type of inferential statistic used to study if there is a statistical difference between two groups. Mathematically, it establishes the problem by assuming that the means of the two distributions are equal (H_0: $\mu_1=\mu_2$). If the t-test rejects the null hypothesis (H_0: $\mu_1=\mu_2$), it indicates that the groups are highly

probably different.

The statistical test can be one-tailed or two-tailed. The **one-tailed test** is appropriate when there is a difference between groups in a specific direction. It is less common than the **two-tailed test.** When choosing a *t* test, you will need to consider two things: whether the groups being compared come from a single population or two different populations, and whether you want to test the difference in a specific direction.

There are three main types of t-test :

- **One Sample t-test :** Compares **mean of a single group against a known**/hypothesized/ population mean.
- **Two Sample: Paired Sample T Test:** Compares means from the **same group at different times.**
- **Two Sample: Independent Sample T Test:** Compares means for **two different groups.**

One Sample t-test:

$$t = \frac{\text{(Sample Mean- Population Mean)}}{\text{Standard Error}}$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$\bar{x}$ Sample mean
μ Population mean
s Sample standard deviation
n Sample size

Two-sample - Paired Sample t-test

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

$\bar{d}$ =Mean of the difference
s=Standard deviation of the difference
n =is the sample size (i.e., size of d)

- If the **calculated t value is less than critical t value or greater that the critical value** (obtained from a critical value table called the T-distribution table) **then reject the null hypothesis.**
- **P-value <significance level (α) => Reject your null hypothesis** in favor of your alternative hypothesis. Your result is statistically significant.
- **P-value >= significance level (α) => Fail to reject your null hypothesis.** Your result is not statistically significant.

Program:

In [1]:
```
import numpy as np
import pandas as pd
from scipy import stats
Reliance_Mart_Data=pd.read_excel(r'C:\Users\Ramya\Desktop\ads sttp data\t val\RelianceDataMart.xlsx')
```

In [2]:
```
Reliance_Mart_Data
```

Out[2]:

	Rice_Bag_Weight
0	24.50
1	24.70
2	25.60
3	25.00
4	24.70
5	23.30
6	23.30
7	24.00
8	25.10
9	24.30
10	23.30
11	24.10
12	24.10
13	24.20
14	25.20
15	24.90
16	24.70
17	24.10
18	25.00
19	24.70
20	24.90
21	25.00
22	24.00
23	23.98
24	24.30
25	24.20
26	24.56
27	24.50
28	24.70

In [3]:
```
Reliance_Mart_Data.describe()
```

Out[3]:

	Rice_Bag_Weight
count	29.000000
mean	24.448207
std	0.569483
min	23.300000
25%	24.100000
50%	24.500000
75%	24.900000
max	25.600000

In [4]:
```
one_sample_result=stats.ttest_1samp(Reliance_Mart_Data,25)
```

In [5]:
```
one_sample_result
```

Out[5]:
```
Ttest_1sampResult(statistic=array([-5.23697685]), pvalue=array([1.45121657e-05]))
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import pandas as pd
from scipy import stats
```

```
prepostscore=pd.read_excel('/content/drive/MyDrive/Colab Notebooks/Pre_Post_Score.xlsx')
```

```
prepostscore
```

	Pre_Score	Post_Score
0	18	22
1	21	25
2	16	17
3	22	24
4	19	16
5	24	29
6	17	20
7	21	23
8	23	19
9	18	20
10	14	15
11	16	15
12	16	18
13	19	26
14	18	18
15	20	24
16	12	18
17	22	25
18	15	19
19	17	16

```
Two_Sample_Results=stats.ttest_rel (prepostscore ["Pre_Score"], prepostscore ["Post_Score"])
```

```
Two_Sample_Results
```

```
Ttest_relResult(statistic=-3.231252665580312, pvalue=0.004394965993185664)
```

2/12/23, 4:54 PM Crocin_Data.ipynb - Colaboratory

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount
```

```
import pandas as pd
from scipy import stats
```

```
Crocin_Data=pd.read_excel('/content/drive/MyDrive/Colab Notebooks/Crocin_Data_ST.xlsx')
```

```
Crocin_Data
```

	Before_Crocin	After_Crocin
0	101.0	99
1	99.0	98
2	101.0	97
3	99.9	99
4	99.8	98
5	98.0	97
6	97.0	99
7	101.0	98
8	102.0	96
9	103.0	98
10	99.0	94
11	99.9	96
12	99.8	97
13	99.7	99
14	101.1	98
15	102.3	97
16	101.0	99
17	99.0	98
18	101.0	97
19	99.9	99
20	99.8	98
21	98.0	96
22	97.0	97
23	101.0	99
24	102.0	97
25	103.0	99
26	99.0	98
27	99.9	97
28	99.8	99

```
Two_Sample_Results=stats.ttest_rel (Crocin_Data ["Before_Crocin"], Crocin_Data ["After_Crocin"])
```

```
Two_Sample_Results
```

```
Ttest_relResult(statistic=7.071712959273876, pvalue=1.0808112658181922e-07)
```

Conclusion: One sample t-test has been done on the reliance data mart dataset and it has been found that difference exists between the rice bag population mean and rice bag sample mean. Two sample paired t-test has been done on the prescore-post score dataset and Crocin dataset. In the prescore-post score dataset difference exists between the mean pre-score before studying the module and mean prescore after studying the module. In the crocin dataset it is found that temperature difference exists before and after having the crocin tablet.

Exercise 1:

Abstract: In this exercise, a paired sample t-test is conducted to evaluate the impact of backend optimization techniques on website response time. Performance measurements collected before and after optimization for the same test cases are analyzed to determine whether the observed reduction in response time is statistically significant. This exercise helps students develop analytical skills in performance evaluation and evidence-based system optimization using inferential statistics.

Dataset:

Website Response Time Optimization Dataset

Test_Case	Response_Time_Before (ms)	Response_Time_After (ms)
T1	420	360
T2	450	390
T3	410	355
T4	470	400
T5	440	380
T6	460	395
T7	430	370
T8	445	385

Exercise 2:

Abstract:

This exercise involves applying a one-sample t-test to analyze daily electricity consumption recorded from smart energy meters. The objective is to examine whether the average energy usage of households significantly differs from a predefined standard consumption level. Students will analyze the data by framing appropriate hypotheses, calculating the test statistic, and interpreting the results to support data-driven decisions related to energy efficiency and consumption monitoring.

Dataset:

Smart Energy Meter Consumption Dataset

Household_ID	**Daily_Energy_Consumption (kWh)**
H1	11.2
H2	10.8
H3	11.5
H4	10.6
H5	11.0
H6	11.3
H7	10.9
H8	11.1
H9	10.7
H10	11.4

(Assumed population mean daily consumption = 11 kWh)